

PARAMETER & PASSWORDFILE

- Create pfile from spfile to location '/home/oracle/init<sid.ora>'

Step 1:

Check the database status

```
SQL> SELECT NAME,OPEN_MODE FROM V$DATABASE;  
  
NAME          OPEN_MODE  
-----  
ORCL          READ WRITE
```

Step 2:

Verifying in which parameter file database is running

```
SQL> SHOW PARAMETER SPFILE;  
  
NAME                                TYPE      VALUE  
-----  
spfile                              string    /u01/app/oracle/product/19.0.0  
                                     /dbhome_1/dbs/spfileorcl.ora
```

Step 3:

Creating the PFILE from SPFILE to location '/home/oracle/init<sid.ora>'

```
SQL> CREATE PFILE='/home/oracle/initorcl.ora' FROM SPFILE;  
  
File created.
```

Step 4:

Now checking PFILE is created or not in OS Level

```
SQL> !ls -ltrh /home/oracle/initorcl.ora  
-rw-r--r--. 1 oracle oinstall 1.1K May 26 19:44 /home/oracle/initorcl.ora
```

- Create parameter file in different location /u01 & start the DB to up and running condition.

Step 1:

Creating the PFILE from SPFILE to location '/u01'

```
SQL> CREATE PFILE='/u01/initorcl.ora' from spfile;  
  
File created.
```

Step 2:

Now checking PFILE is created or not in OS Level

```
SQL> !ls -ltrh /u01/initrcl.ora  
-rw-r--r--. 1 oracle oinstall 1.1K May 26 20:00 /u01/initrcl.ora
```

Step 3:

Now Starting the database from '/u01' location

```
SQL> STARTUP PFILE='/u01/initrcl.ora';  
ORACLE instance started.  
  
Total System Global Area 520091504 bytes  
Fixed Size 8898416 bytes  
Variable Size 243269632 bytes  
Database Buffers 264241152 bytes  
Redo Buffers 3682304 bytes  
Database mounted.  
Database opened.
```

Step 4:

Database is Started Using PFILE

```
SQL>  
SQL> show parameter spfile;  
  
NAME                                TYPE      VALUE  
-----                                -  
spfile                              string  
SQL>
```

Step 5:

Checking the database condition

```
SQL> select name,open_mode from v$database;  
  
NAME      OPEN_MODE  
-----  
ORCL      READ WRITE
```

➤ How to check all parameters associated with database?

In Oracle Database DBA, you can check all database parameters using the following commands:

1. SQL> show parameter;

```
SQL> show parameter;
```

NAME	TYPE	VALUE
DBFIPS_140	boolean	FALSE
active_instance_count	integer	
add_account_info_tracking	string	LOCAL
add_redirect_dml	boolean	FALSE
allow_global_dblinks	boolean	FALSE
allow_group_access_to_sga	boolean	FALSE
allow_rowid_column_type	boolean	FALSE
approx_for_aggregation	boolean	FALSE
approx_for_count_distinct	boolean	FALSE
approx_for_percentile	string	none
aq_tm_processes	integer	1

NAME	TYPE	VALUE
archive_lag_target	integer	0
asm_diskstring	string	
asm_preferred_read_failure_groups	string	
audit_file_dest	string	/u01/app/oracle/admin/orcl/adu
audit_sys_operations	boolean	TRUE
audit_syslog_level	string	mp
audit_trail	string	DB
autotask_max_active_pdb	integer	2
awr_pdb_autoflush_enabled	boolean	FALSE
awr_pdb_max_parallel_slaves	integer	10

2. SQL> show parameter spfile;

```
SQL> SHOW PARAMETER SPFILE;
```

NAME	TYPE	VALUE
spfile	string	/u01/app/oracle/product/19.0.0/ /dbhome_1/dbs/spfileorcl.ora

3. SQL> show parameter memory;

```
SQL> show parameter memory;
```

NAME	TYPE	VALUE
hi_shared_memory_address	integer	0
inmemory_adg_enabled	boolean	TRUE
inmemory_automatic_level	string	OFF
inmemory_clause_default	string	
inmemory_expressions_usage	string	ENABLE
inmemory_force	string	DEFAULT
inmemory_max_populate_servers	integer	0
inmemory_optimized_arithmetic	string	DISABLE
inmemory_prefer_xmem_memcompress	string	
inmemory_prefer_xmem_priority	string	
inmemory_query	string	ENABLE
inmemory_size	big integer	0
inmemory_trickle_repopulate_servers_percent	integer	1
inmemory_virtual_columns	string	MANUAL
inmemory_xmem_size	big integer	0
memory_max_target	big integer	0
memory_target	big integer	0
optimizer_inmemory_aware	boolean	TRUE
shared_memory_address	integer	0

4. SQL> show parameter sga;

```
SQL> SHOW PARAMETER SGA;
```

NAME	TYPE	VALUE
allow_group_access_to_sga	boolean	FALSE
lock_sga	boolean	FALSE
pre_page_sga	boolean	TRUE
sga_max_size	big integer	496M
sga_min_size	big integer	0
sga_target	big integer	496M
unified_audit_sga_queue_size	integer	1048576

5. SQL> show parameter pga;

```
SQL> SHOW PARAMETER PGA;
```

NAME	TYPE	VALUE
pga_aggregate_limit	big integer	2G
pga_aggregate_target	big integer	166M

6. SQL> show parameter undo;

```
SQL> SHOW PARAMETER UNDO;
```

NAME	TYPE	VALUE
temp_undo_enabled	boolean	FALSE
undo_management	string	AUTO
undo_retention	integer	900
undo_tablespace	string	UNDOTBS1

7. SQL> show parameter archive;

```
SQL> SHOW PARAMETER ARCHIVE;
```

NAME	TYPE	VALUE
archive_lag_target	integer	0
log_archive_config	string	
log_archive_dest	string	
log_archive_dest_1	string	
log_archive_dest_10	string	
log_archive_dest_11	string	
log_archive_dest_12	string	
log_archive_dest_13	string	
log_archive_dest_14	string	

8. SQL> select num,name,value,issys_modifiable from v\$parameter fetch first 20 rows only;

```
SQL> SELECT NUM,NAME,VALUE,ISSYS_MODIFIABLE FROM V$PARAMETER FETCH FIRST 20 ROWS ONLY;
```

NUM	NAME	VALUE	ISSYS_MODIFIABLE
52	lock_name_space		FALSE
53	processes	300	FALSE
54	sessions	472	IMMEDIATE
71	timed_statistics	TRUE	IMMEDIATE
72	timed_os_statistics	0	IMMEDIATE
73	resource_limit	TRUE	IMMEDIATE
74	license_max_sessions	0	IMMEDIATE
75	license_sessions_warning	0	IMMEDIATE
84	long_module_action	TRUE	IMMEDIATE
106	standby_db_preserve_states	NONE	FALSE
110	instance_abort_delay_time	0	IMMEDIATE
125	cpu_count	1	IMMEDIATE
126	cpu_min_count	1	IMMEDIATE
136	instance_groups		FALSE
145	event		FALSE
160	sga_max_size	520093696	FALSE
167	use_large_pages	TRUE	FALSE
176	pre_page_sga	TRUE	FALSE
177	shared_memory_address	0	FALSE
178	hi_shared_memory_address	0	FALSE

20 rows selected.

VALUE	MEANING
IMMEDIATE	Change takes effect immediately
FALSE	Database restart required
DEFERRED	Change affects only new sessions

➤ How to change specific parameters with Scope=both/spfile/memory?

Scope	Effect
MEMORY	Current running instance only
SPFILE	Permanent change after restart
BOTH	Current memory + SPFILE

Step 1:

Checking the impact on the database when scope = spfile

```
SQL> SELECT NAME,OPEN_MODE FROM V$DATABASE;
```

NAME	OPEN_MODE
ORCL	READ WRITE

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	900

```
SQL> ALTER SYSTEM SET UNDO_RETENTION= 1500 SCOPE=SPFILE;
```

System altered.

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	900

```
SQL> SHU IMMEDIATE
Database closed.
Database dismounted.
ORACLE instance shut down.
```

```
SQL>
```

```
SQL>
```

```
SQL> STARTUP
```

```
ORACLE instance started.
```

```
Total System Global Area  520091504 bytes
Fixed Size                  8898416 bytes
Variable Size               243269632 bytes
Database Buffers            264241152 bytes
Redo Buffers                 3682304 bytes
```

```
Database mounted.
```

```
Database opened.
```

```
SQL>
```

```
SQL>
```

```
SQL>
```

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	1500

Step 2:

Checking the impact on the database when scope = memory

```
SQL> ALTER SYSTEM SET UNDO_RETENTION= 100 SCOPE=MEMORY;
```

System altered.

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	100

```
SQL>
```

```
SQL>
```

```
SQL> SELECT NAME,OPEN_MODE FROM V$DATABASE;
```

NAME	OPEN_MODE
ORCL	READ WRITE

```
SQL> SHU IMMEDIATE
```

Database **closed**.

Database dismounted.

ORACLE instance shut down.

```
SQL>
```

```
SQL>
```

```
SQL> STARTUP
```

ORACLE instance started.

Total System Global Area 520091504 bytes

Fixed Size 8898416 bytes

Variable Size 243269632 bytes

Database Buffers 264241152 bytes

Redo Buffers 3682304 bytes

Database mounted.

Database opened.

```
SQL>
```

```
SQL>
```

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	1500

Step 3:

Checking the impact on the database when scope = both

```
SQL> SELECT NAME,OPEN_MODE FROM V$DATABASE;
```

NAME	OPEN_MODE
ORCL	READ WRITE

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	1500

```
SQL>
```

```
SQL>
```

```
SQL> ALTER SYSTEM SET UNDO_RETENTION= 900 scope=both;
```

System altered.

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	900

```

SQL> shu immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL> startup
ORACLE instance started.

Total System Global Area  520091504 bytes
Fixed Size                  8898416 bytes
Variable Size              243269632 bytes
Database Buffers           264241152 bytes
Redo Buffers                3682304 bytes
Database mounted.
Database opened.
SQL>
SQL>
SQL> SHOW PARAMETER UNDO_RETENTION;

```

NAME	TYPE	VALUE
-----	-----	-----
undo_retention	integer	900

- **How to check which parameter files required DB bounce for the changes to get reflected?**

Static Parameters → Require database restart (DB bounce)

If V\$PARAMETER, ISSYS_MODIFIABLE value is false then need to restart the database

```

SQL> SELECT NAME,VALUE,ISSYS_MODIFIABLE FROM V$PARAMETER WHERE NAME='processes';

NAME                                VALUE                                ISSYS_MODIFIABLE
-----                                -
processes                           300                                FALSE

SQL> ALTER SYSTEM SET processes=500 SCOPE=MEMORY;
ALTER SYSTEM SET processes=500 SCOPE=MEMORY
*
ERROR at line 1:
ORA-02095: specified initialization parameter cannot be modified

SQL> ALTER SYSTEM SET processes=500 SCOPE=SPFILE;

System altered.

SQL> SELECT NAME,VALUE,ISSYS_MODIFIABLE FROM V$PARAMETER WHERE NAME='processes';

NAME                                VALUE                                ISSYS_MODIFIABLE
-----                                -
processes                           300                                FALSE

```

```

SQL> SHU IMMEDIATE
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL> STARTUP
ORACLE instance started.

Total System Global Area  520091504 bytes
Fixed Size                  8898416 bytes
Variable Size              205520896 bytes
Database Buffers           301989888 bytes
Redo Buffers                3682304 bytes
Database mounted.
Database opened.
SQL>
SQL>
SQL> SELECT NAME,VALUE,ISSYS_MODIFIABLE FROM V$PARAMETER WHERE NAME='processes';

NAME                                VALUE                                ISSYS_MODIFIABLE
-----                                -
processes                           500                                FALSE

```

➤ How to check which parameter files which can be change dynamically?

Dynamic parameter -> Not require database restart

If V\$PARAMETER, ISSYS_MODIFIABLE value is immediate then need not to restart the database.

```
SQL> SELECT NAME,VALUE,ISSYS_MODIFIABLE FROM V$PARAMETER WHERE NAME='undo_retention';
```

NAME	VALUE	ISSYS_MODIFIABLE
undo_retention	1100	IMMEDIATE

```
SQL> alter system set undo_retention=300 scope=memory;
```

System altered.

```
SQL> SELECT NAME,VALUE,ISSYS_MODIFIABLE FROM V$PARAMETER WHERE NAME='undo_retention';
```

NAME	VALUE	ISSYS_MODIFIABLE
undo_retention	300	IMMEDIATE

➤ How to check which parameter files cannot be changed?

In Oracle DBA, if the ISSYS_MODIFIABLE column value is FALSE, the parameter is considered static and cannot be modified dynamically for the current running instance.

```
SQL> select NUM,NAME,ISSES_MODIFIABLE from v$parameter where ISSES_MODIFIABLE='FALSE' fetch first 5 rows only;
```

NUM	NAME	ISSES_MODIFIABLE
52	lock_name_space	FALSE
53	processes	FALSE
54	sessions	FALSE
73	resource_limit	FALSE
74	license_max_sessions	FALSE

➤ Create password file with sys password as oracle1

Step 1:

Checking the password file value with the below command

```
SQL> SHOW PARAMETER remote_login_passwordfile;
```

Common values:

- EXCLUSIVE
 - SHARED
 - NONE
-
- ✓ **EXCLUSIVE** → One database, full password file management
 - ✓ **SHARED** → Shared password file, limited modifications
 - ✓ **NONE** → No password file used


```
SQL> SHOW PARAMETER REMOTE_LOGIN_PASSWORDFILE;
```

NAME	TYPE	VALUE
remote_login_passwordfile	string	EXCLUSIVE

```
[oracle@localhost dbs]$ orapwd file=orapworcl password=oracle1 entries=10
```

Explanation:

- file=orapwORCL → Name of the password file
- password=oracle1 → SYS user password
- entries=10 → Number of users allowed in the password file

```
[oracle@localhost dbs]$ ll
total 10524
-rw-r--r--. 1 oracle oinstall 1034 May  9 13:04 createdb.sql
-rw-rw----. 1 oracle oinstall 1544 May 28 20:18 hc_orcl.dat
-rw-rw----. 1 oracle oinstall 1544 May 22 15:21 hc_prodc.dat
-rwxrwxr-x. 1 oracle oinstall 1544 May 27 14:51 hc_prod.dat
-rw-rw----. 1 oracle oinstall 1544 May 18 11:20 hc_sha.dat
-rw-rw----. 1 oracle oinstall 1544 May 18 13:54 hc_SHA.dat
-rwxrwxr-x. 1 oracle oinstall 3079 May 14 2015 init.ora
-rw-r--r--. 1 oracle oinstall 1116 May 21 18:47 initorcl.ora
-rwxrwxr-x. 1 oracle oinstall 1137 May 22 15:20 initprod.ora
-rw-rw----. 1 oracle oinstall 24 May  2 17:04 lkORCL
-rwxrwxr-x. 1 oracle oinstall 24 May  2 15:09 lkPROD
-rw-rw----. 1 oracle oinstall 24 May 18 11:26 lkSHA
-rw-rw----. 1 oracle oinstall 2048 May  2 17:06 orapworcl
-rw-rw----. 1 oracle oinstall 2048 May 28 21:38 orapworcl12
-rw-rw----. 1 oracle oinstall 2048 May 18 11:29 orapwSHA
-rw-rw----. 1 oracle oinstall 10698752 May 27 15:54 snapcf_orcl.f
-rw-rw----. 1 oracle oinstall 3584 May 28 20:18 spfileorcl.ora
-rwxrwxr-x. 1 oracle oinstall 3584 May 27 14:51 spfileprod.ora
-rw-rw----. 1 oracle oinstall 3584 May 18 13:51 spfileSHA.ora
```